

Gabriel Mendes Bonato

Full-Stack Software Engineer · Industrial IoT & Energy

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Araraquara, São Paulo, Brazil · Remote (GMT-3) · English (fluent) · Portuguese (native)

SUMMARY

Full-stack software engineer with 3 years in industrial IoT — factory-device systems, ERP/PoS, and now utility-scale solar. Lead engineer of MeuWatt, a monitoring and remote-control platform running across a fleet of utility-scale solar plants: device protocols, backend and time-series data, frontend, and infrastructure. Also the platform's operator: fleet deploys, production incident response, and on-site commissioning.

EXPERIENCE

Software Engineer — Industrial IoT & Energy, MeuWatt

2025 – Present

- Lead engineering on a 3-tier IoT platform — on-site edge collectors, cloud API, web dashboard — monitoring a fleet of utility-scale solar plants in production.
- Built the Python edge collector and its multi-vendor device layer: Modbus TCP polling with register maps and drivers for inverters (Sungrow, Huawei, GoodWe), protection relays, thermal relays, weather stations, energy meters and RS-485 gateways, plus an automatic fault-detection state machine.
- Built the cloud API on FastAPI and TimescaleDB: fleet-wide telemetry ingestion and an idempotent, replay-safe energy-loss attribution engine; traced a production database cost overrun to unindexed read paths rather than write volume and returned spend to baseline with targeted indexes and query rewrites.
- Building an AI analytics layer on fleet telemetry: structured raw time-series into training-ready datasets and developing ML/LLM pipelines for automated operational insights.
- Built a secure remote-operations rail — audited command queue, WebSocket + HTTP transports, TOTP 2FA step-up — covering configuration and power state for inverters, protection relays, thermal relays and multi-vendor RS-485 gateways, so routine intervention no longer requires sending someone to the plant.
- Built the React + TypeScript operations dashboard: real-time monitoring, generation and availability reporting, breakdown management, and fleet-health administration.
- Run the delivery pipeline from commit to running plant: automated tests (pytest), CI image builds (GitHub Actions), cloud and on-site fleet deployments (Docker, Ansible), and centralized logging.

Full-Stack / Mobile Developer, JN Moura — Araraquara

2024 – 2025

- Developed and maintained a Flutter ERP with Point-of-Sale (PoS) integrations; diagnosed and fixed data-processing defects.
- Extended a React application that processes university entrance-exam candidate lists.
- Automated Git workflows on GitLab with custom scripts, reducing merge errors and improving team throughput; worked in Scrum (sprints, dailies, planning).

Front-End Developer, Fabrisense

2023 – 2024

- Built dashboards and device-monitoring views in React for industrial IoT factory equipment; developed components and integrated REST endpoints.

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Dart

Backend & Data: FastAPI, SQLAlchemy/Alembic, REST APIs, WebSockets, OAuth2/JWT, TOTP 2FA, PostgreSQL, TimescaleDB, time-series pipelines, pytest

Frontend & Mobile: React 19, Vite, SPA architecture, dashboards & charting, Flutter

Infra / DevOps: Docker, Ansible, GitHub Actions, CI/CD, Vercel, Render, Graylog, Linux, GitLab

AI / ML: dataset preparation & time-series feature engineering, LLM integration, prompt & context engineering, AI agents & agentic workflows, Model Context Protocol (MCP)

Industrial: Modbus TCP, solar inverters (Sungrow / Huawei / GoodWe), protection relays, energy meters, SCADA-style monitoring